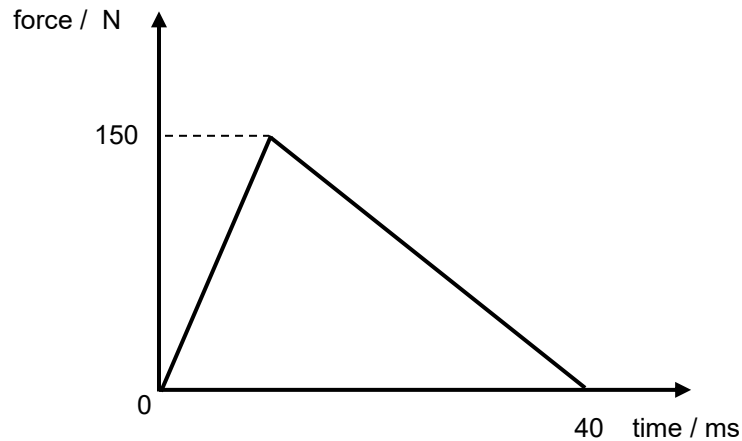


- 5 A tennis ball of mass 0.060 kg is moving normally towards a racket with an initial velocity of 5.0 m s^{-1} and rebounds in the opposite direction.

The force of the racket exerted on the tennis ball varies with time as shown.



What is the final velocity of the ball when it leaves the racket?

- | | | | | | | | |
|----------|-----------------------|----------|-----------------------|----------|-----------------------|----------|-----------------------|
| A | 45 m s^{-1} | B | 50 m s^{-1} | C | 55 m s^{-1} | D | 95 m s^{-1} |
|----------|-----------------------|----------|-----------------------|----------|-----------------------|----------|-----------------------|

Answer: A

Take final velocity direction as positive direction.

Area under the force-time graph represents the impulse on the tennis ball.

$$F\Delta t = \Delta p$$

$$\frac{1}{2}(0.040)(150) = m\Delta v$$

$$3 = 0.060 (v_f - (-5.0))$$

$$v_f = 45 \text{ m s}^{-1}$$

6

A tower is pulled by a bundle of steel cables anchored to the ground, as shown by the